

PROOF OF LEMMA 1

Proof.

$$\begin{aligned}
& \|w_t - w_{t-\tau,j}^i\|^2 = \|w_t - w_{t-\tau} + w_{t-\tau} - w_{t-\tau,j}^i\|^2 \\
& \leq 2\|w_t - w_{t-\tau}\|^2 + 2\|w_{t-\tau} - w_{t-\tau,j}^i\|^2 \\
& \leq 2\gamma^2 \left\| \sum_{l=1}^{\tau} \sum_{k=1}^K \nabla f(w_{t+l-\tau_l,k-1}^i) \right\|^2 + 2\gamma^2 \left\| \sum_{l=1}^j \nabla f(w_{t-\tau,l}^i) \right\|^2 \\
& \leq 2\gamma^2 \tau K \sum_{l=1}^{\tau} \sum_{k=1}^K \|\nabla f(w_{t+l-\tau_l,k-1}^i)\|^2 \\
& + 2\gamma^2 j \sum_{l=1}^j \|\nabla f(w_{t-\tau,l}^i)\|^2 \\
& \leq 2\gamma^2 \tau K \sum_{l=1}^{\tau} \sum_{k=1}^K G^2 + 2\gamma^2 j \sum_{l=1}^j G^2 \\
& \leq 2\gamma^2 \tau^2 K^2 G^2 + 2\gamma^2 j^2 G^2, \tag{1}
\end{aligned}$$

which completes the proof. $\square$

PROOF OF THEOREM 2

We provide the proof of the main convergence theorem (Theorem 2 in the main paper) as below.

Proof. According to L -smooth of the loss function in Assumption 1, we have

$$\begin{aligned}
& F(w_{t+1}) \\
& \leq F(w_t) - \underbrace{\gamma \langle \nabla F(w_t), G_{t-\tau}^i \rangle}_{T_1} + \underbrace{\frac{\gamma^2}{2} \|G_{t-\tau}^i\|_2^2}_{T_2}. \tag{2}
\end{aligned}$$

In the following, we tend to show that the L_2 norm of the gradient converges to 0. To do this, we first present the bound of term T_1 and then the bound of term T_2 . Our analysis relies on the expectation of whether the worker receiving the global model from the server. We assume that the duration between two updates of global model in the server follows the uniform distribution, i.e., $\text{time}(w_{t-1}, w_t) \sim U$ for any t . Similarly, we consider the computation time of each local iteration follows the uniform distribution. Since each worker i computes K local iterations between receiving the global model $w_{t-\tau}$ and uploading update G^i to the server to update the global model w_t , we consider that the worker has probability receiving τ global models. Specifically, considering the uniformity of the global update and local update, each worker i receives the global model with the probability r in each k -th local iteration where $k\% \lfloor K/\tau \rfloor = 0$ when $\tau \leq K$. In this paper, we mainly discuss the case $\tau \leq K$ which can be easily extended to the case when $\tau > K$. Definitely, the number of combination of whether receiving global models is $2^{\tau-1}$. For analysis, in each local iteration $k\% \lfloor K/\tau \rfloor = 0$, we denote $\tilde{w}_{t-\tau+l,j}^i$ by the previous local model without receiving the global model, and $\nabla F_i(w_{t-\tau+l,j}^i)$ by its gradient. Now, under this case, we take expectation over whether receiving the global model by using $\mathbb{E}_{t_s}$. Besides, we take expectation $\mathbb{E}_{z,i,\tau,t_s}$ of independent

items including client i , samples z , and the staleness τ . For simplicity, we use $\mathbb{E}$ to denote $\mathbb{E}_{z,i,\tau,t_s}$ obtaining:

$$\begin{aligned}
\mathbb{E}T_1 &= \mathbb{E} \left\langle \nabla F(w_t), \sum_{k=1}^K \nabla f(w_{t-\tau,k-1}^i) \right\rangle \\
&= \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \left[\sum_{j=1}^{\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right. \right. \\
&\quad \left. \left. + \sum_{l=1}^{\tau-1} \sum_{j=1}^{\lfloor K/\tau \rfloor} ((1-r)\nabla F_i(\tilde{w}_{t-\tau+l,j}^i) + r\nabla F_i(w_{t-\tau+l,j}^i)) \right] \right\rangle \\
&\geq \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \left[\sum_{j=1}^{2\lfloor K/\tau \rfloor} (1-r)\nabla F_i(w_{t-\tau,j}^i) \right. \right. \\
&\quad \left. \left. + r \sum_{l=1}^{\tau-1} \sum_{j=1}^{\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right] \right\rangle \\
&\geq (1-r) \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\rangle \\
&\quad + r \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \sum_{l=1}^{\tau-1} \sum_{j=1}^{\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right\rangle, \tag{3}
\end{aligned}$$

Since

$$\begin{aligned}
& \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\rangle \\
&= 2\lfloor K/\tau \rfloor \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\rangle \\
&= \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- \left\| \nabla F(w_t) - \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \Big] \\
&= \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_t) \right\|^2 \\
&- \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \|\nabla F_i(w_t) - \nabla F_i(w_{t-\tau,j}^i)\|^2 \Big] \\
&\geq_{(a)} \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \|\nabla F_i(w_t) - \nabla F_i(w_{t-\tau,j}^i)\|^2 \Big] \\
&\geq \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- \frac{L}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \|w_t - w_{t-\tau,j}^i\|^2 \Big] \\
&\geq \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- \frac{L}{n} \sum_{i=1}^n \frac{1}{\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \gamma^2 G^2 (\tau^2 K^2 + j^2) \Big] \\
&\geq \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- 2L\gamma^2 G^2 (\tau^2 K^2 + \lfloor K/\tau \rfloor^2) \Big], \tag{4}
\end{aligned}$$

where (a) is due to the Cauchy-Schwarz inequality. The last second inequality is due to Lemma 1. Similarly,

$$\begin{aligned}
& \mathbb{E} \left\langle \nabla F(w_t), \frac{1}{n} \sum_{i=1}^n \sum_{l=1}^{\tau-1} \sum_{j=1}^{\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right\rangle \\
&\geq \frac{\tau \lfloor K/\tau \rfloor}{4} \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{\tau} \sum_{l=1}^{\tau-1} \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right\|^2 \\
&- 2L\gamma^2 G^2 (\tau^2 K^2 + 4\lfloor K/\tau \rfloor^2) \Big], \tag{5}
\end{aligned}$$

Bringing (4) and (5) back to (3) derives the bound of T_1 :

$$\begin{aligned}
\mathbb{E} T_1 &\geq (1-r) \lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&- 2L\gamma^2 G^2 (\tau^2 K^2 + \lfloor K/\tau \rfloor^2) \Big] \\
&+ \frac{r\tau \lfloor K/\tau \rfloor}{4} \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&+ \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{\tau} \sum_{l=1}^{\tau-1} \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right\|^2 \\
&- 2L\gamma^2 G^2 (\tau^2 K^2 + 4\lfloor K/\tau \rfloor^2) \Big], \tag{6}
\end{aligned}$$

Now, we turn back to T_2 , and derive that

$$\begin{aligned}
\mathbb{E} \|G_{t-\tau}^i\|_2^2 &= \mathbb{E} \frac{1}{n} \sum_{i=1}^n \left\| \sum_{k=1}^K \nabla f(w_{t-\tau,k-1}^i) \right\|_2^2 \\
&\leq \mathbb{E} \frac{K}{n} \sum_{i=1}^n \sum_{k=1}^K \left\| \nabla f(w_{t-\tau,k-1}^i) - \nabla F_i(w_{t-\tau,k-1}^i) \right. \\
&\quad \left. + \nabla F_i(w_{t-\tau,k-1}^i) \right\|_2^2 \\
&\leq 2\mathbb{E} \frac{K}{n} \sum_{i=1}^n \sum_{k=1}^K \left\| \nabla f(w_{t-\tau,k-1}^i) - \nabla F_i(w_{t-\tau,k-1}^i) \right\|_2^2 \\
&+ 2\mathbb{E} \frac{K}{n} \sum_{i=1}^n \sum_{k=1}^K \left\| \nabla F_i(w_{t-\tau,k-1}^i) \right\|_2^2 \\
&\leq 2K^2 \sigma^2 + 2K^2 G^2 \tag{7}
\end{aligned}$$

Substituting (6) and (7) back into (2) yields

$$\begin{aligned}
F(w_{t+1}) &\leq F(w_t) - \gamma(1-r)\lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&\quad + \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau,j}^i) \right\|^2 \\
&\quad - 2L\gamma^2 G^2(\tau^2 K^2 + \lfloor K/\tau \rfloor^2) \Big] \\
&\quad - \gamma \frac{r\tau \lfloor K/\tau \rfloor}{4} \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&\quad + \left\| \frac{1}{n} \sum_{i=1}^n \frac{1}{\tau} \sum_{l=1}^{\tau-1} \frac{1}{2\lfloor K/\tau \rfloor} \sum_{j=1}^{2\lfloor K/\tau \rfloor} \nabla F_i(w_{t-\tau+l,j}^i) \right\|^2 \\
&\quad - 2L\gamma^2 G^2(\tau^2 K^2 + 4\lfloor K/\tau \rfloor^2) \Big] \\
&\quad + \frac{\gamma^2}{2}(2K^2\sigma^2 + 2K^2G^2) \\
&\leq F(w_t) - \gamma(1-r)\lfloor K/\tau \rfloor \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&\quad - 2L\gamma^2 G^2(\tau^2 K^2 + \lfloor K/\tau \rfloor^2) \Big] \\
&\quad - \gamma \frac{r\tau \lfloor K/\tau \rfloor}{4} \left[\mathbb{E} \|\nabla F(w_t)\|^2 \right. \\
&\quad - 2L\gamma^2 G^2(\tau^2 K^2 + 4\lfloor K/\tau \rfloor^2) \Big] \\
&\quad + \gamma^2(K^2\sigma^2 + K^2G^2). \tag{8}
\end{aligned}$$

Re-organizing the (8) by moving $\mathbb{E} \|\nabla F(w_t)\|^2$ to the left hand of the inequality derives

$$\begin{aligned}
&\lfloor K/\tau \rfloor \gamma (1-r + \frac{r\tau}{4}) \mathbb{E} \|\nabla F(w_t)\|^2 \\
&\leq F(w_t) - F(w_{t+1}) \\
&\quad + 2\gamma^3(1-r)\lfloor K/\tau \rfloor LG^2(\tau^2 K^2 + \lfloor K/\tau \rfloor^2) \\
&\quad + \gamma^3 \frac{r\tau \lfloor K/\tau \rfloor}{2} LG^2(\tau^2 K^2 + 4\lfloor K/\tau \rfloor^2) \\
&\quad + \gamma^2(K^2\sigma^2 + K^2G^2) \tag{9}
\end{aligned}$$

For simplicity, we consider $\lfloor K/\tau \rfloor$ is an integer and thus, we have

$$\frac{rK\gamma}{4} \leq \lfloor K/\tau \rfloor \gamma (1-r + \frac{r\tau}{4}). \tag{10}$$

Besides, considering the minima of the loss function as $F_* \leq F(w_t)$ and summarizing both sides of (9) from $t = 1$ to T and divided by $\frac{rK\gamma}{4}$ obtains

$$\begin{aligned}
\frac{1}{T} \sum_{t=1}^T \|\nabla F(w_t)\|^2 &\leq \frac{4(F(w_1) - F_*)}{rK\gamma T} \\
&\quad + \frac{8\gamma^2(1-r)}{r} LG^2(\tau_{max} K^2 + K) \\
&\quad + 2\gamma^2 LG^2(\tau_{max}^2 K^2 + 4K^2) + \frac{4\gamma}{r}(K\sigma^2 + KG^2),
\end{aligned}$$

which completes the proof. $\square$

Algorithm 1: UCB Algorithm for Model Broadcasting

Input: Broadcast time parameters m and b , upperbound index γ , time between consecutive broadcast Δt , and training loss: $loss$

- 1 **Initialization:** N arms, total number of actions $T \geq N$,
- 2 Action list $A = [a_1, a_2, \dots, a_N]$, Pull times: $t = 0$
- 3 **while** $t < T$ **do**
- 4 **if** broadcast is *True* **then**
- 5 **if** $t < N$ **then**
- 6 $I_t = t$
- 7 **else**
- 8 **for each action** $a_i \in A$ **do**
- 9 $\hat{u}_{i,t-1} = t_{i,t-1}/n_{i,t-1}$
- 10 $UCB_{i,t-1} = \hat{u}_{i,t-1} + \sqrt{\frac{2 \cdot \log t}{n_{i,t-1}}}$
- 11 $I_t = \operatorname{argmax}_{i \in \{1 \dots N\}} UCB_{i,t-1}$
- 12 $t_{\text{downtime}} = m \cdot a_{I_t} + b$
- 13 $n_{I_t,t} = 1$
- 14 $t_{I_t,t} = e_t^{-loss} \cdot \gamma / \Delta t$
- 15 $t = t + 1$

CONVERGENCE WITH $\gamma = 1/\sqrt{T}$

The following corollary of Theorem 2 characterizes the convergence rate under a decaying learning rate.

Corollary 1. *Considering the problem in the main paper under Assumption 1, setting the learning rate $\gamma = \frac{1}{\sqrt{T}}$, the convergence rate of the proposed BAP algorithm is*

$$\begin{aligned}
\frac{1}{T} \sum_{t=1}^T \|\nabla F(w_t)\|^2 &\leq \frac{4(F(w_1) - F_*)}{rK\sqrt{T}} \\
&\quad + \frac{8(1-r)}{rT} LG^2(\tau_{max} K^2 + K) \\
&\quad + \frac{2LG^2(\tau_{max}^2 K^2 + 4K^2)}{T} + \frac{4}{r\sqrt{T}}(K\sigma^2 + KG^2),
\end{aligned}$$

where F_* is the minima of the loss function and τ_{max} is the maximum of the staleness.

Remark 1. Corollary 2 shows that BAP converges to 0 with a sub-linear rate $O(\frac{1}{\sqrt{T}})$ when the learning rate $\gamma = \frac{1}{\sqrt{T}}$, which is the same as sequential SGD.

THE UCB BROADCASTING ALGORITHM

Algorithm 1 details the UCB procedure for elastic model broadcasting discussed in the main paper. The algorithm maintains, for each arm a_i , the accumulated reward $t_{i,t-1}$ and the pull count $n_{i,t-1}$, and selects the arm with the highest upper confidence bound at each broadcast.

EXTENDED RELATED WORK

Distributed Edge Learning

DEL has emerged as a promising paradigm for training machine learning models in edge environments [1], [2]. This approach harnesses the computational resources of distributed edge devices while keeping raw data localized, ensuring both privacy preservation and reduced communication overhead.

Federated learning (FL) [3], [4] has been a cornerstone in this domain, providing a decentralized training framework where edge devices collaboratively train a global model by exchanging only model updates. However, traditional FL methods assume stable communication links and homogeneous device capabilities, which are rarely guaranteed in real-world edge networks.

To address these limitations, researchers have proposed various adaptations that enhance the flexibility, robustness, and scalability of DEL systems. For example, hierarchical federated learning frameworks introduce an intermediate layer between edge devices and the central server, leveraging edge servers to perform local aggregation before transmitting updates to the central server [5]–[7]. This multi-tiered architecture not only reduces communication latency but also minimizes bandwidth consumption and server overload. Other works focus on improving the security and privacy of DEL [8], [9]. Bonawitz et al. propose robust encryption mechanisms and differential privacy techniques to protect sensitive information during the training process [10]. These methods ensure that raw data remains on local devices, and model updates are securely aggregated, mitigating the risk of data breaches and inference attacks. Recent advancements also address dynamic and heterogeneous edge environments, where devices vary in computational power, energy resources, and network stability. Techniques like adaptive learning rate adjustments [11], [12], device clustering based on capabilities [13], [14], and selective update aggregation [15], [16] have been explored to enhance the robustness of federated systems. These adaptations enable the system to dynamically allocate tasks, optimize resource utilization, and ensure fair participation of all devices, regardless of their constraints.

Synchronization Schemes

Synchronization schemes play a crucial role in the efficiency and performance of DEL frameworks. Traditional synchronous training methods, where all edge workers complete their computations and update the central model in lockstep, are widely used due to their simplicity and robust convergence guarantees [3], [17]. However, these methods often suffer from the straggler problem as slow or unreliable devices can delay the entire training process. To address this, asynchronous schemes have been extensively studied. Asynchronous Stochastic Gradient Descent (ASGD) is a well-known approach that allows devices to update the global model independently, without waiting for others [18]. Although ASGD alleviates the straggler issue, it introduces challenges related to gradient staleness, which can adversely impact the convergence rate. To mitigate this, staleness-aware approaches dynamically adjust the learning rate or apply correction factors to reduce the effect of outdated gradients [19], [20].

Hybrid synchronization mechanism is a promising solution for DEL to combine the advantages of both paradigms for balancing convergence efficiency and robustness to device heterogeneity. In [21], [22], edge devices are grouped into smaller clusters to perform local synchronization, followed by periodic

global updates. This reduces the impact of slower devices while ensuring a reasonable degree of model consistency. While these approaches effectively solve the problem of heterogeneity, they still compromise convergence rates due to increased staleness in updates. In contrast, our method combines the benefits of asynchronization with efficient broadcast mechanisms for model updating, achieving both reduced overhead and high convergence rates, making it well-suited for dynamic edge environments.

Rateless Code

Rateless codes, such as LT codes [23] and Raptor codes [24], are a class of erasure codes that dynamically generate an indefinite number of encoded packets from a finite source data block. These codes are particularly attractive in scenarios with unpredictable or lossy communication channels, as successful decoding is possible once a sufficient number of encoded packets are received, regardless of which specific packets arrive. Rateless codes have been extensively used in wireless communication systems to address packet loss and improve data reliability without requiring retransmissions [25]–[27]. Their ability to adapt to varying channel conditions and eliminate the need for feedback makes them highly efficient for large-scale data distribution in resource-constrained environments.

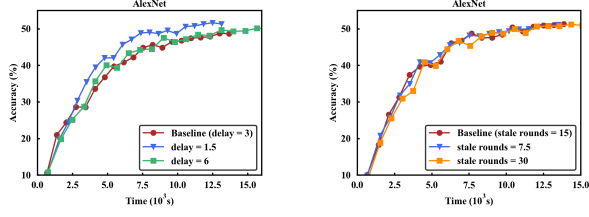
We aim to leverage rateless codes in DEL to optimize the communication efficiency of model broadcasting. In conventional broadcast schemes, packet loss caused by dynamic wireless channel conditions often requires retransmissions, leading to significant latency [28], [29]. Rateless coding addresses this challenge by enabling edge devices to decode the global model upon receiving a sufficient number of encoded packets, irrespective of which specific packets are received. This approach effectively reduces the time required for model dissemination and enhances the overall efficiency of the broadcast process.

ADDITIONAL EXPERIMENTS

A. Impact of Straggler Settings

Stragglers, representing workers with delayed responses in distributed systems, are a critical factor in real-world scenarios. To investigate the influence of stragglers, we consider two factors of stragglers: relay time (delay) and stale rounds, as shown in Figure 1. Relay time refers to the additional delay introduced by slower workers. In Figure 1(a), as the delay multiplier increases, the training process generally becomes less stable. However, the BAP algorithm demonstrates notable robustness, particularly when the delay exceeds 3 times the baseline. This indicates that BAP is capable of mitigating the impact of extreme delays by adaptively avoiding reliance on workers with excessive delays.

Stale rounds represent the number of iterations a straggling worker's gradient update lags behind the most recent state of the model. In Figure 1(b), longer stale rounds can disrupt synchronization and hinder the convergence process. However, the results reveal that BAP is highly resilient to variations in stale round settings. Whether the stale round value is reduced



(a) Multiplier of the straggler (b) Occupying rounds of each straggler

Fig. 1: Experiment results from BAP with different straggler settings.

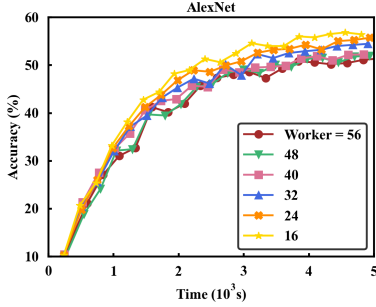
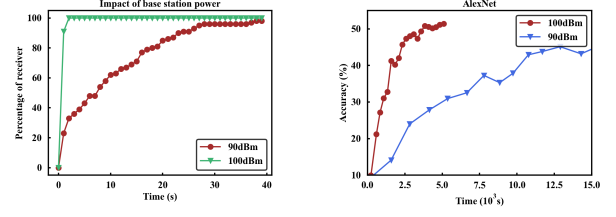


Fig. 2: Experiment results from BAP with different numbers of workers.

to 7.5 or increased to 30, the convergence speed remains nearly unchanged. This suggests that BAP effectively handles outdated updates without compromising overall performance, making it robust to asynchronous communication and delayed updates in distributed learning environments.

B. Scalability

Figure 2 illustrates the influence of varying the number of workers under consistent environmental conditions, using AlexNet as the backbone network in conjunction with the BAP algorithm. The results indicate that a configuration with 16 workers achieves the best overall performance, striking an optimal balance between computational efficiency and communication overhead. This setup ensures sufficient parallelism for distributed learning while minimizing delays caused by synchronization and data transfer among workers. However, as the number of workers increases beyond 16, the performance begins to degrade slightly. This decline can be attributed to the growing communication costs and synchronization delays associated with managing larger worker pools. Although increasing the number of workers theoretically enhances computational power, the system experiences diminishing returns as the overhead of gradient aggregation, data partitioning, and coordination outweighs the benefits of additional parallelism. Specifically, configurations with 32 or more workers introduce significant communication burdens, which hinder convergence speed and overall performance.



(a) Data transfer time during a single broadcast (b) Training accuracy vs. time across base station power settings

Fig. 3: Experiment results from BAP with different base station power.

C. Impact of Base Station Power

Figure 3 illustrates the impact of base station power on the efficiency of data transmission and the resulting training performance in a distributed learning scenario. Two base station settings are compared: one operating at 90dBm and the other at 100dBm.

The first graph (Figure 3(a)) shows the percentage of receivers completing the data transfer during a single broadcast. It is evident that the base station power significantly influences the data transmission speed. The base station with 100dBm power delivers the entire broadcast within approximately 2 seconds, achieving a rapid and efficient data transfer. In contrast, the 90dBm base station requires as much as 40 seconds to complete the same broadcast. This stark difference highlights the critical role of base station power in determining the speed and reliability of communication.

The second graph (Figure 3(b)) illustrates the relationship between model accuracy and training time under the two base station configurations. With 100dBm power, the model achieves the desired accuracy in roughly one-third of the time compared to the 90dBm setting. The faster convergence in the 100dBm scenario is directly linked to the reduced communication delays, which enable more frequent and timely updates to the model. Conversely, the slower data transfer in the 90dBm case leads to extended training times, as workers must wait longer to synchronize their gradients and updates.

These results underscore the significant influence of base station power on the overall efficiency of distributed learning systems. In real-world scenarios, where communication channels are more dynamic and susceptible to fluctuations, the challenges posed by lower power levels can become even more pronounced. This highlights the necessity of adaptive broadcast strategies that can dynamically adjust to varying channel conditions and optimize data transfer processes.

D. Additional Results on ResNet-18

Figure 4 reports additional training curves of BAP on ResNet-18 with CIFAR-10 under different numbers of workers ($K = 8$), complementing the scalability study in the main paper.

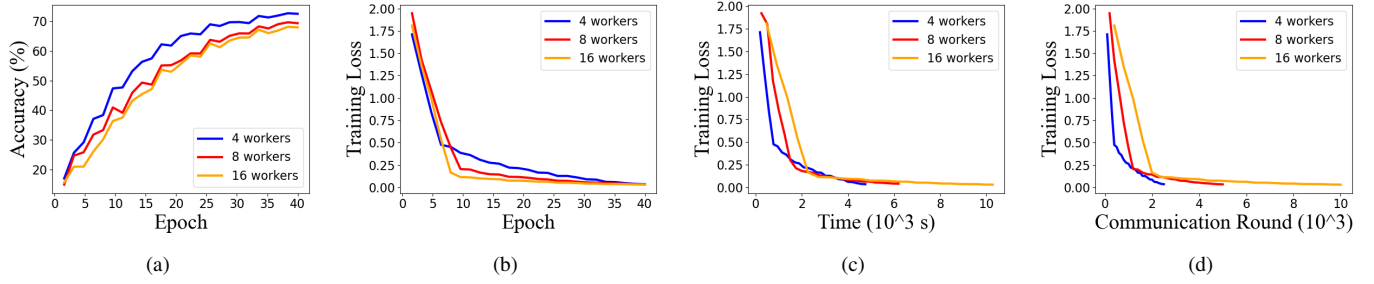


Fig. 4: Experiment results showing different workers from ResNet-18 on CIFAR-10 with $K = 8$.

REFERENCES

- [1] M. Threadgill and A. Gerstlauer, "A Survey of Distributed Learning in Cloud, Mobile, and Edge Settings," *arXiv preprint arXiv:2405.15079*, 2024.
- [2] J. Zhang, Z. Qu, C. Chen, H. Wang, Y. Zhan, B. Ye, and S. Guo, "Edge Learning: The Enabling Technology for Distributed Big Data Analytics in the Edge," *ACM Comput. Surv.*, vol. 54, no. 7, Jul. 2021. [Online]. Available: <https://doi.org/10.1145/3464419>
- [3] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*. PMLR, 2017, pp. 1273–1282.
- [4] J. Wu, F. Dong, H. Leung, Z. Zhu, J. Zhou, and S. Drew, "Topology-aware federated learning in edge computing: A comprehensive survey," *ACM Computing Surveys*, vol. 56, no. 10, pp. 1–41, 2024.
- [5] L. Liu, J. Zhang, S. Song, and K. B. Letaief, "Hierarchical federated learning with quantization: Convergence analysis and system design," *IEEE Transactions on Wireless Communications*, vol. 22, no. 1, pp. 2–18, 2022.
- [6] Q. Ma, Y. Xu, H. Xu, J. Liu, and L. Huang, "FedUC: A unified clustering approach for hierarchical federated learning," *IEEE Transactions on Mobile Computing*, 2024.
- [7] X. Liu, Z. Yan, Y. Zhou, D. Wu, X. Chen, and J. H. Wang, "Optimizing parameter mixing under constrained communications in parallel federated learning," *IEEE/ACM Transactions on Networking*, vol. 31, no. 6, pp. 2640–2652, 2023.
- [8] J. Chen, H. Yan, Z. Liu, M. Zhang, H. Xiong, and S. Yu, "When federated learning meets privacy-preserving computation," *ACM Computing Surveys*, vol. 56, no. 12, pp. 1–36, 2024.
- [9] A. Asheralieva and D. Niyato, "Secure and efficient coded multi-access edge computing with generalized graph neural networks," *IEEE Transactions on Mobile Computing*, vol. 22, no. 9, pp. 5504–5524, 2022.
- [10] K. Bonawitz, H. Eichner, W. Grieskamp, D. Huba, A. Ingerman, V. Ivanov, C. Kiddon, J. Konečný, S. Mazzocchi, H. B. McMahan *et al.*, "Towards federated learning at scale: System design," *arXiv preprint arXiv:1902.01046*, 2019.
- [11] M. P. Uddin, Y. Xiang, B. Cai, X. Lu, J. Yearwood, and L. Gao, "Arfl: Adaptive and robust federated learning," *IEEE Transactions on Mobile Computing*, 2023.
- [12] S. Zheng, W. Yuan, X. Wang, and L. Duan, "Adaptive Federated Learning via New Entropy Approach," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 11920–11936, Dec. 2024. [Online]. Available: <https://doi.ieeecomputersociety.org/10.1109/TMC.2024.3402080>
- [13] B. Gong, T. Xing, Z. Liu, W. Xi, and X. Chen, "Towards Hierarchical Clustered Federated Learning With Model Stability on Mobile Devices," *IEEE Transactions on Mobile Computing*, vol. 23, no. 6, p. 7148–7164, Nov. 2023. [Online]. Available: <https://doi.org/10.1109/TMC.2023.3332637>
- [14] M. Wazzeah, M. Arafah, H. Sami, H. Ould-Slimane, C. Talhi, A. Mourad, and H. Otrok, "CRSFL: Cluster-based Resource-aware Split Federated Learning for Continuous Authentication," *arXiv preprint arXiv:2405.07174*, 2024.
- [15] H. Zhou, M. Li, P. Sun, B. Guo, and Z. Yu, "Accelerating Federated Learning via Parameter Selection and Pre-Synchronization in Mobile Edge-Cloud Networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 11, pp. 10313–10328, 2024.
- [16] W. Wu, L. He, W. Lin, and C. Maple, "FedProf: Selective Federated Learning Based on Distributional Representation Profiling," *IEEE Trans. Parallel Distrib. Syst.*, vol. 34, no. 6, p. 1942–1953, Jun. 2023. [Online]. Available: <https://doi.org/10.1109/TPDS.2023.3265588>
- [17] T. Cheatham, A. Fahmy, D. Stefanescu, and L. Valiant, "Bulk synchronous parallel computing—a paradigm for transportable software," *Tools and Environments for Parallel and Distributed Systems*, pp. 61–76, 1996.
- [18] K. Mishchenko, F. Bach, M. Even, and B. E. Woodworth, "Asynchronous SGD beats minibatch SGD under arbitrary delays," *Advances in Neural Information Processing Systems*, vol. 35, pp. 420–433, 2022.
- [19] M. Yu, J. Choi, J. Lee, and S. Oh, "Staleness aware semi-asynchronous federated learning," *Journal of Parallel and Distributed Computing*, vol. 193, p. 104950, 2024.
- [20] Z. Wang, Z. Zhang, Y. Tian, Q. Yang, H. Shan, W. Wang, and T. Q. S. Quek, "Asynchronous Federated Learning Over Wireless Communication Networks," *IEEE Transactions on Wireless Communications*, vol. 21, no. 9, pp. 6961–6978, 2022.
- [21] Z. Li, C. Huang, K. Gai, Z. Lu, J. Wu, L. Chen, Y. Xu, and K.-K. R. Choo, "AsyFed: Accelerated Federated Learning With Asynchronous Communication Mechanism," *IEEE Internet of Things Journal*, vol. 10, no. 10, pp. 8670–8683, 2023.
- [22] S. Sun, Z. Zhang, Q. Pan, M. Liu, Y. Wang, T. He, Y. Chen, and Z. Wu, "Staleness-Controlled Asynchronous Federated Learning: Accuracy and Efficiency Tradeoff," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 12621–12634, 2024.
- [23] M. Luby, "LT codes," in *The 43rd Annual IEEE Symposium on Foundations of Computer Science, 2002. Proceedings*. IEEE Computer Society, 2002, pp. 271–271.
- [24] A. Shokrollahi, "Raptor codes," *IEEE transactions on information theory*, vol. 52, no. 6, pp. 2551–2567, 2006.
- [25] D. L. Romero, N. B. Chang, and A. R. Margetts, "Practical non-binary rateless codes for wireless channels," in *2013 Asilomar Conference on Signals, Systems and Computers*. IEEE, 2013, pp. 2084–2088.
- [26] P. Schulz, A. Traßl, A. N. Barreto, and G. Fettweis, "Efficient and reliable wireless communications via multi-connectivity using rateless codes in single-and multi-user scenarios," *IEEE Transactions on Wireless Communications*, vol. 20, no. 9, pp. 5714–5729, 2021.
- [27] J. Lv, G. Chen, and W. Dong, "Exploiting rateless codes and cross-layer optimization for low-power wide-area networks," *ACM Transactions on Sensor Networks*, vol. 18, no. 4, pp. 1–24, 2023.
- [28] H. Tian, H. Zhang, J. Jia, M. Dong, and K. Ota, "FedBroadcast: Exploit broadcast channel for fast convergence in wireless federated learning," *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4652–4663, 2022.
- [29] H. Rydén, R. Moosavi, and E. G. Larsson, "Federated Learning with Multi-resolution Model Broadcast," *arXiv preprint arXiv:2405.19886*, 2024.